Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc. I.T Semester V)

Applied Business Analytics

Practical – VI

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Class: TYIT	Batch: 1
Date of Assignment: 02/09/2026	Date/Time of Submission: 02/09/2026

6. Decision Making under Uncertainty:

a. Develop a decision tree model to make business decisions considering uncertainties and associated probabilities at each decision point.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("supermarket_sales.csv")

# Q1. Display first 10 records
print("\nQ1. First 10 Records:")
print(df.head(10))

# Create revenue column using Total
df["revenue"] = df["Total"]

# Q2. Dataset information
print("\nQ2. Dataset Information:")
print("Number of Records:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("Column Names:")
print(df.columns.tolist())
print("\nData Types:")
print(df.dtypes)

# Q3. Unique product lines
print("\nQ3. Unique Product Lines:")
print(df["Product line"].unique())

# Q4. Total revenue by product line
total_revenue = df.groupby("Product line")["revenue"].sum()
print("\nQ4. Total Revenue by Product Line:")
print(total_revenue)

# Q5. Average revenue by product line
```

```
# Q5. Average revenue by product line
average_revenue = df.groupby("Product line")["revenue"].mean()
print("\nQ5. Average Revenue by Product Line:")
print(average_revenue)

# Q6. Total quantity sold by product line
total_quantity = df.groupby("Product line")["Quantity"].sum()
print("\nQ6. Total Quantity Sold:")
print(total_quantity)

# Q7. Median revenue
median_revenue = df["revenue"].median()
print("\nQ7. Median Revenue:", median_revenue)

# Q8. Classify transactions
df["revenue_level"] = df["revenue"].apply(
    lambda x: "High Revenue" if x >= median_revenue else "Low Revenue"
)

# Q9. Display revenue_level
print("\nQ9. Revenue Level:")
print(df[["revenue", "revenue_level"]].head(10))

# Q10. Number of High/Low transactions
count_table = pd.crosstab(df["Product line"], df["revenue_level"])
print("\nQ10. High and Low Revenue Transactions:")
print(count_table)

# Q11. Probability of High/Low Revenue
probability_table = count_table.div(count_table.sum(axis=1), axis=0)
print("\nQ11. Revenue Probabilities:")
print(probability_table)
```

```python
# Q12. Verify probabilities add to 1
probability_sum = probability_table.sum(axis=1)
print("\nQ12. Probability Sum:")
print(probability_sum)

# Q13. Average revenue for High/Low Revenue
payoff_table = df.groupby(
    ["Product line", "revenue_level"]
)["revenue"].mean().unstack()

print("\nQ13. Average Revenue / Payoff:")
print(payoff_table)

# Q14. Create DataFrame with Product Line, Revenue Level,
# Probability and Payoff
decision_data = []

for product in probability_table.index:
    for level in probability_table.columns:
        decision_data.append([
            product,
            level,
            probability_table.loc[product, level],
            payoff_table.loc[product, level]
        ])

decision_df = pd.DataFrame(
    decision_data,
    columns=["Product Line", "Revenue Level", "Probability", "Payoff"]
)

print("\nQ14. Decision DataFrame:")
print(decision_df)
```

```python
# Q15. Decision alternatives and uncertain outcomes
print("\nQ15. Decision Alternatives:")
print(probability_table.index.tolist())

print("\nUncertain Outcomes:")
print(probability_table.columns.tolist())

# Q16 & Q17. Decision Tree Data
print("\nQ16-Q17. Decision Tree:")
for product in probability_table.index:
    print("\nDecision:", product)
    for level in probability_table.columns:
        print(
            "  ->", level,
            "| Probability:", round(probability_table.loc[product, level], 4),
            "| Payoff:", round(payoff_table.loc[product, level], 2)
        )

# Q18. Total Revenue Visualization
plt.figure(figsize=(10, 6))
total_revenue.sort_values().plot(kind="barh")
plt.title("Total Revenue by Product Line")
plt.xlabel("Total Revenue")
plt.ylabel("Product Line")
plt.tight_layout()
plt.show()
```

```python
# Q19. Probability Visualization
probability_table.plot(
    kind="bar",
    figsize=(10, 6)
)
plt.title("Probability of High and Low Revenue")
plt.xlabel("Product Line")
plt.ylabel("Probability")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Q20. Highest probability of High Revenue
highest_high_probability = probability_table["High Revenue"].idxmax()
highest_probability = probability_table["High Revenue"].max()

print("\nQ20. Highest Probability of High Revenue:")
print(highest_high_probability, "=", round(highest_probability, 4))
```

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Output:

```
Q1. First 10 Records:
    Invoice ID Branch       City Customer type  Gender  \
0  750-67-8428      A     Yangon        Member  Female
1  226-31-3081      C  Naypyitaw        Normal  Female
2  631-41-3108      A     Yangon        Normal    Male
3  123-19-1176      A     Yangon        Member    Male
4  373-73-7910      A     Yangon        Normal    Male
5  699-14-3026      C  Naypyitaw        Normal    Male
6  355-53-5943      A     Yangon        Member  Female
7  315-22-5665      C  Naypyitaw        Normal  Female
8  665-32-9167      A     Yangon        Member  Female
9  692-92-5582      B   Mandalay        Member  Female

             Product line  Unit price  Quantity   Tax 5%     Total       Date  \
0       Health and beauty       74.69         7  26.1415  548.9715   1/5/2019
1  Electronic accessories       15.28         5   3.8200   80.2200   3/8/2019
2      Home and lifestyle       46.33         7  16.2155  340.5255   3/3/2019
3       Health and beauty       58.22         8  23.2880  489.0480  1/27/2019
4       Sports and travel       86.31         7  30.2085  634.3785   2/8/2019
5  Electronic accessories       85.39         7  29.8865  627.6165  3/25/2019
6  Electronic accessories       68.84         6  20.6520  433.6920  2/25/2019
7      Home and lifestyle       73.56        10  36.7800  772.3800  2/24/2019
8       Health and beauty       36.26         2   3.6260   76.1460  1/10/2019
9      Food and beverages       54.84         3   8.2260  172.7460  2/20/2019
```

```
Q2. Dataset Information:
Number of Records: 1000
Number of Columns: 18
Column Names:
['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line',

Data Types:
Invoice ID                     str
Branch                         str
City                           str
Customer type                  str
Gender                         str
Product line                   str
Unit price                 float64
Quantity                     int64
Tax 5%                     float64
Total                      float64
Date                           str
Time                           str
Payment                        str
cogs                       float64
gross margin percentage    float64
gross income               float64
Rating                     float64
revenue                    float64
dtype: object
```

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```
Q3. Unique Product Lines:
<StringArray>
[     'Health and beauty', 'Electronic accessories',     'Home and lifestyle',
      'Sports and travel',     'Food and beverages',    'Fashion accessories']
Length: 6, dtype: str

Q4. Total Revenue by Product Line:
Product line
Electronic accessories    54337.5315
Fashion accessories       54305.8950
Food and beverages        56144.8440
Health and beauty         49193.7390
Home and lifestyle        53861.9130
Sports and travel         55122.8265
Name: revenue, dtype: float64

Q5. Average Revenue by Product Line:
Product line
Electronic accessories    319.632538
Fashion accessories       305.089298
Food and beverages        322.671517
Health and beauty         323.643020
Home and lifestyle        336.636956
Sports and travel         332.065220
Name: revenue, dtype: float64
```

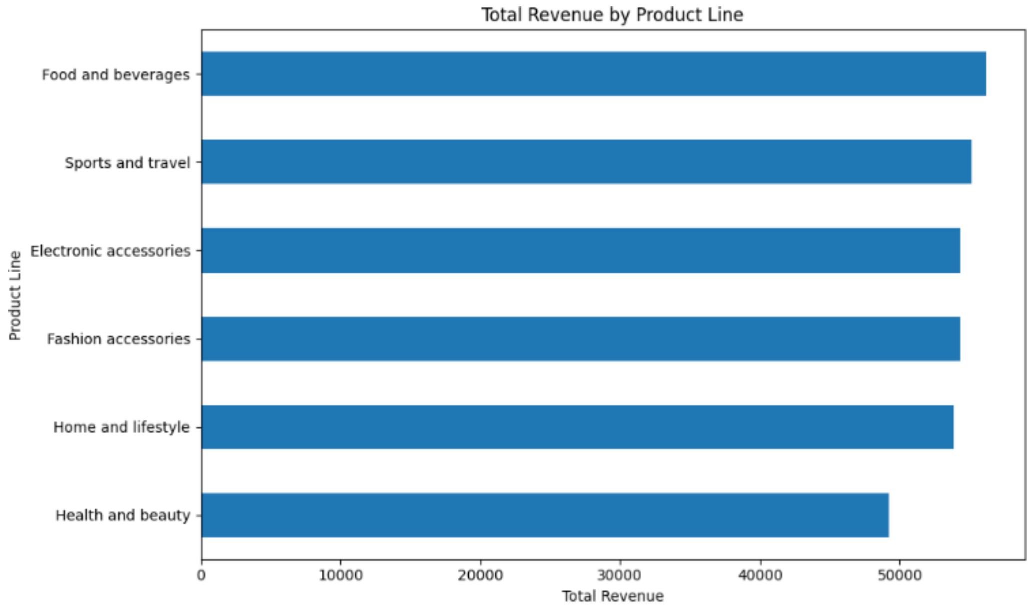
```
Q6. Total Quantity Sold:
Product line
Electronic accessories    971
Fashion accessories       902
Food and beverages        952
Health and beauty         854
Home and lifestyle        911
Sports and travel         920
Name: Quantity, dtype: int64

Q7. Median Revenue: 253.848

Q9. Revenue Level:
   revenue revenue_level
0  548.9715  High Revenue
1   80.2200   Low Revenue
2  340.5255  High Revenue
3  489.0480  High Revenue
4  634.3785  High Revenue
5  627.6165  High Revenue
6  433.6920  High Revenue
7  772.3800  High Revenue
8   76.1460   Low Revenue
```

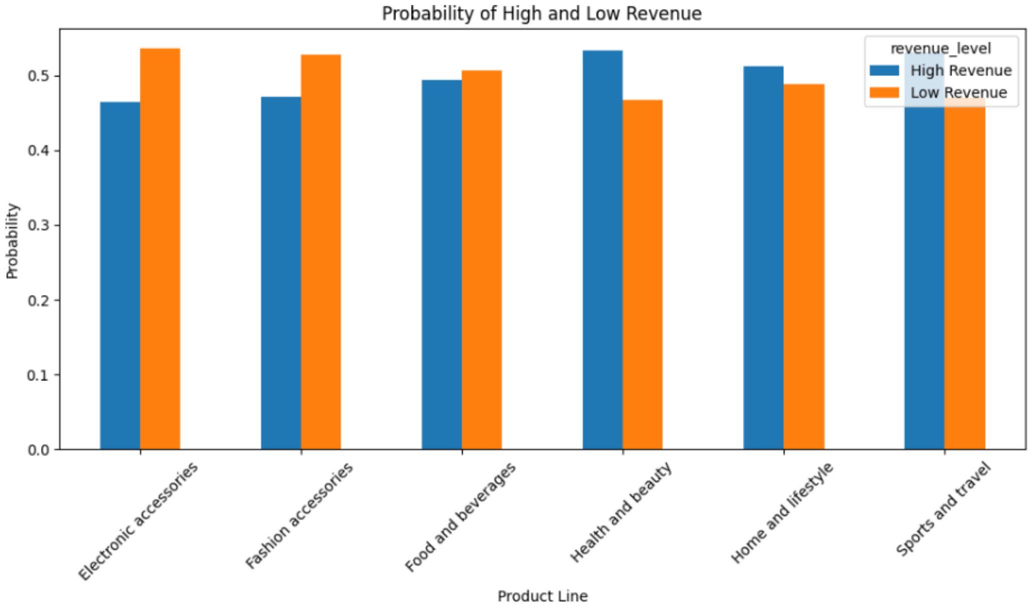
```
Q10. High and Low Revenue Transactions:
revenue_level            High Revenue  Low Revenue
Product line
Electronic accessories             79           91
Fashion accessories                84           94
Food and beverages                 86           88
Health and beauty                  81           71
Home and lifestyle                 82           78
Sports and travel                  88           78

Q11. Revenue Probabilities:
revenue_level            High Revenue  Low Revenue
Product line
Electronic accessories       0.464706     0.535294
Fashion accessories          0.471910     0.528090
Food and beverages           0.494253     0.505747
Health and beauty            0.532895     0.467105
Home and lifestyle           0.512500     0.487500
Sports and travel            0.530120     0.469880
```

```
Q12. Probability Sum:
Product line
Electronic accessories    1.0
Fashion accessories       1.0
Food and beverages        1.0
Health and beauty         1.0
Home and lifestyle        1.0
Sports and travel         1.0
dtype: float64

Q13. Average Revenue / Payoff:
revenue_level           High Revenue  Low Revenue
Product line
Electronic accessories    537.006152   130.923577
Fashion accessories       512.329375   119.896037
Food and beverages        515.127680   134.589358
Health and beauty         497.221407   125.616972
Home and lifestyle        529.582738   133.796519
Sports and travel         518.546438   121.676154
```

```
Q14. Decision DataFrame:
              Product Line Revenue Level  Probability      Payoff
0   Electronic accessories  High Revenue     0.464706  537.006152
1   Electronic accessories   Low Revenue     0.535294  130.923577
2      Fashion accessories  High Revenue     0.471910  512.329375
3      Fashion accessories   Low Revenue     0.528090  119.896037
4       Food and beverages  High Revenue     0.494253  515.127680
5       Food and beverages   Low Revenue     0.505747  134.589358
6        Health and beauty  High Revenue     0.532895  497.221407
7        Health and beauty   Low Revenue     0.467105  125.616972
8       Home and lifestyle  High Revenue     0.512500  529.582738
9       Home and lifestyle   Low Revenue     0.487500  133.796519
10       Sports and travel  High Revenue     0.530120  518.546438
11       Sports and travel   Low Revenue     0.469880  121.676154

Q15. Decision Alternatives:
['Electronic accessories', 'Fashion accessories', 'Food and beverages',
```

```
Uncertain Outcomes:
['High Revenue', 'Low Revenue']

Q16-Q17. Decision Tree:

Decision: Electronic accessories
  -> High Revenue | Probability: 0.4647 | Payoff: 537.01
  -> Low Revenue | Probability: 0.5353 | Payoff: 130.92

Decision: Fashion accessories
  -> High Revenue | Probability: 0.4719 | Payoff: 512.33
  -> Low Revenue | Probability: 0.5281 | Payoff: 119.9

Decision: Food and beverages
  -> High Revenue | Probability: 0.4943 | Payoff: 515.13
  -> Low Revenue | Probability: 0.5057 | Payoff: 134.59
```






```
Q20. Highest Probability of High Revenue:
Health and beauty = 0.5329
```

b. Apply the concept of expected value to evaluate different decision alternatives and select the optimal one.

Code:

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("supermarket_sales.csv")

# Use Total as Revenue
df["revenue"] = df["Total"]

# Calculate median revenue
median_revenue = df["revenue"].median()

# Classify revenue
df["revenue_level"] = df["revenue"].apply(
    lambda x: "High Revenue" if x >= median_revenue else "Low Revenue"
)

# Calculate probabilities
count_table = pd.crosstab(df["Product line"], df["revenue_level"])
probability_table = count_table.div(count_table.sum(axis=1), axis=0)

# Calculate payoffs
payoff_table = df.groupby(
    ["Product line", "revenue_level"]
)["revenue"].mean().unstack()

# PART B: APPLY EXPECTED VALUE

# Q21. Explain Expected Value
print("\nQ21. Expected Value:")
print("Expected Value is used to compare different product lines.")
print("The product line with the highest Expected Value is selected.")
```

```python
# Q22. Formula
print("\nQ22. Formula:")
print("EV = (P(High Revenue) × High Revenue Payoff) +")
print("     (P(Low Revenue) × Low Revenue Payoff)")

# Q23. Calculate Expected Value of one product line
product = probability_table.index[0]

high_probability = probability_table.loc[product, "High Revenue"]
low_probability = probability_table.loc[product, "Low Revenue"]

high_payoff = payoff_table.loc[product, "High Revenue"]
low_payoff = payoff_table.loc[product, "Low Revenue"]

ev_one = (high_probability * high_payoff) + \
         (low_probability * low_payoff)

print("\nQ23. Expected Value of", product, ":")
print(round(ev_one, 2))

# Q24. Calculate Expected Value for every product line
expected_values = (
    probability_table["High Revenue"] * payoff_table["High Revenue"]
    + probability_table["Low Revenue"] * payoff_table["Low Revenue"]
)

print("\nQ24. Expected Value for Every Product Line:")
print(expected_values.round(2))
```

```python
# Q25. Create Expected Value DataFrame
ev_df = pd.DataFrame({
    "Product Line": probability_table.index,
    "High Revenue Probability":
        probability_table["High Revenue"].values,
    "High Revenue Payoff":
        payoff_table["High Revenue"].values,
    "Low Revenue Probability":
        probability_table["Low Revenue"].values,
    "Low Revenue Payoff":
        payoff_table["Low Revenue"].values,
    "Expected Value":
        expected_values.values
})

print("\nQ25. Expected Value DataFrame:")
print(ev_df.round(2))

# Q26. Compare Expected Values
print("\nQ26. Comparison of Expected Values:")
print(
    ev_df[["Product Line", "Expected Value"]]
    .sort_values("Expected Value", ascending=False)
    .round(2)
)
```

```python
# Q27. Highest Expected Value
highest_ev = ev_df.loc[ev_df["Expected Value"].idxmax()]

print("\nQ27. Highest Expected Value:")
print("Product Line:", highest_ev["Product Line"])
print("Expected Value:", round(highest_ev["Expected Value"], 2))

# Q28. Lowest Expected Value
lowest_ev = ev_df.loc[ev_df["Expected Value"].idxmin()]

print("\nQ28. Lowest Expected Value:")
print("Product Line:", lowest_ev["Product Line"])
print("Expected Value:", round(lowest_ev["Expected Value"], 2))

# Q29. Rank Product Lines
ranking = ev_df.sort_values(
    "Expected Value",
    ascending=False
).reset_index(drop=True)

ranking["Rank"] = ranking.index + 1

print("\nQ29. Ranking:")
print(
    ranking[
        ["Rank", "Product Line", "Expected Value"]
    ].round(2)
)
```

```python
# Q30. Bar Chart of Expected Values
plt.figure(figsize=(10, 6))
plt.bar(
    ev_df["Product Line"],
    ev_df["Expected Value"]
)
plt.title("Expected Value of Each Product Line")
plt.xlabel("Product Line")
plt.ylabel("Expected Value")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Q31. Compare highest High Revenue probability
highest_probability_product = probability_table[
    "High Revenue"
].idxmax()

highest_ev_product = expected_values.idxmax()

print("\nQ31. Comparison:")
print(
    "Highest probability of High Revenue:",
    highest_probability_product
)
print(
    "Highest Expected Value:",
    highest_ev_product
)

# Q32. Are they the same?
print("\nQ32. Result:")
```

```python
if highest_probability_product == highest_ev_product:
    print(
        "Yes, both are the same product line."
    )
else:
    print(
        "No, they are different product lines."
    )

# Q33. Change probability and recalculate EV
product = highest_ev_product

old_high_probability = probability_table.loc[
    product, "High Revenue"
]

new_high_probability = old_high_probability + 0.10

if new_high_probability > 1:
    new_high_probability = 1

new_low_probability = 1 - new_high_probability

new_ev_probability = (
    new_high_probability *
    payoff_table.loc[product, "High Revenue"]
    +
    new_low_probability *
    payoff_table.loc[product, "Low Revenue"]
)

print("\nQ33. Changed Probability:")
print("Product Line:", product)
```

```python
print(
    "New High Revenue Probability:",
    round(new_high_probability, 4)
)
print(
    "New Low Revenue Probability:",
    round(new_low_probability, 4)
)
print(
    "New Expected Value:",
    round(new_ev_probability, 2)
)

# Q34. Change payoff and recalculate EV
old_high_payoff = payoff_table.loc[
    product, "High Revenue"
]

new_high_payoff = old_high_payoff + 50

new_ev_payoff = (
    probability_table.loc[product, "High Revenue"]
    * new_high_payoff
    +
    probability_table.loc[product, "Low Revenue"]
    * payoff_table.loc[product, "Low Revenue"]
)

print("\nQ34. Changed Payoff:")
print("Product Line:", product)
print(
    "New High Revenue Payoff:",
    round(new_high_payoff, 2)
)
```

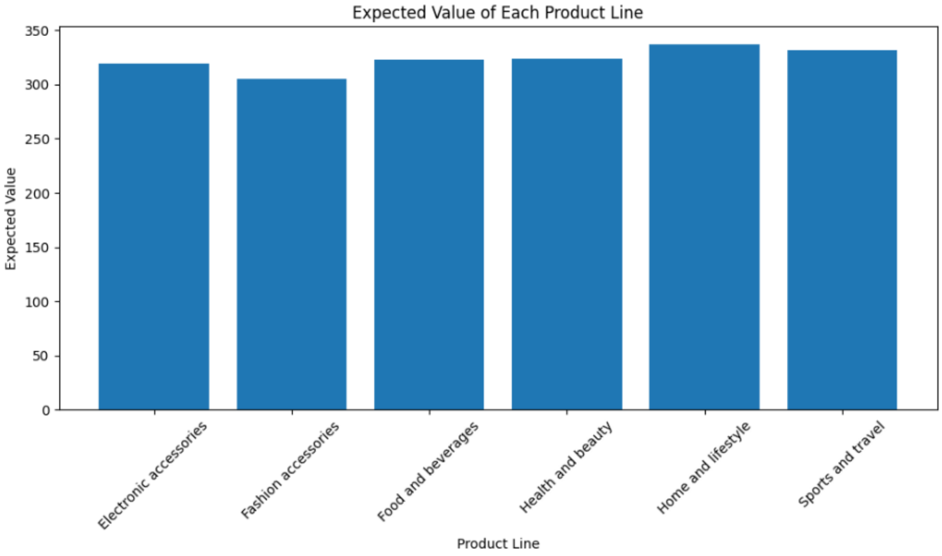
```python
print(
    "New Expected Value:",
    round(new_ev_payoff, 2)
)

# Q35. Automatically select product with highest EV
best_product = expected_values.idxmax()
best_ev = expected_values.max()

print("\nQ35. Automatic Decision:")
print("Selected Product Line:", best_product)
print("Highest Expected Value:", round(best_ev, 2))

# Q36. Final Decision Tree with Expected Value
print("\nQ36. Final Decision Tree:")

for product in probability_table.index:

    print("\n", product)

    print(
        " ├── High Revenue | Probability:",
        round(
            probability_table.loc[
                product, "High Revenue"
            ], 4
        ),
        "| Payoff:",
        round(
            payoff_table.loc[
                product, "High Revenue"
            ], 2
        )
```

```python
            ], 2
        )
    )

    print(
        "  └─ Low Revenue  | Probability:",
        round(
            probability_table.loc[
                product, "Low Revenue"
            ], 4
        ),
        "| Payoff:",
        round(
            payoff_table.loc[
                product, "Low Revenue"
            ], 2
        )
    )

    print(
        "  Expected Value:",
        round(expected_values.loc[product], 2)
    )

# Q37. Business Recommendation
print("\nQ37. Business Recommendation:")

print(
    "The recommended product line is",
    best_product,
    "because it has the highest Expected Value of",
    round(best_ev, 2),
    "."
)
```
✓ 0.2s

Output:

```
Q21. Expected Value:
Expected Value is used to compare different product lines.
The product line with the highest Expected Value is selected.

Q22. Formula:
EV = (P(High Revenue) × High Revenue Payoff) +
     (P(Low Revenue) × Low Revenue Payoff)

Q23. Expected Value of Electronic accessories :
319.63

Q24. Expected Value for Every Product Line:
Product line
Electronic accessories     319.63
Fashion accessories        305.09
Food and beverages         322.67
Health and beauty          323.64
Home and lifestyle         336.64
Sports and travel          332.07
dtype: float64

Q25. Expected Value DataFrame:
             Product Line  High Revenue Probability  High Revenue Payoff
0  Electronic accessories                      0.46               537.01
1     Fashion accessories                      0.47               512.33
2      Food and beverages                      0.49               515.13
3       Health and beauty                      0.53               497.22
4      Home and lifestyle                      0.51               529.58
5       Sports and travel                      0.53               518.55
```

```
Q26. Comparison of Expected Values:
             Product Line  Expected Value
4      Home and lifestyle          336.64
5       Sports and travel          332.07
3       Health and beauty          323.64
2      Food and beverages          322.67
0  Electronic accessories          319.63
1     Fashion accessories          305.09

Q27. Highest Expected Value:
Product Line: Home and lifestyle
Expected Value: 336.64

Q28. Lowest Expected Value:
Product Line: Fashion accessories
Expected Value: 305.09

Q29. Ranking:
   Rank            Product Line  Expected Value
0     1      Home and lifestyle          336.64
1     2       Sports and travel          332.07
2     3       Health and beauty          323.64
3     4      Food and beverages          322.67
4     5  Electronic accessories          319.63
5     6     Fashion accessories          305.09
```




```
Q31. Comparison:
Highest probability of High Revenue: Health and beauty
Highest Expected Value: Home and lifestyle

Q32. Result:
No, they are different product lines.

Q33. Changed Probability:
Product Line: Home and lifestyle
New High Revenue Probability: 0.6125
New Low Revenue Probability: 0.3875
New Expected Value: 376.22

Q34. Changed Payoff:
Product Line: Home and lifestyle
New High Revenue Payoff: 579.58
New Expected Value: 362.26

Q35. Automatic Decision:
Selected Product Line: Home and lifestyle
Highest Expected Value: 336.64

Q36. Final Decision Tree:

 Electronic accessories
  ├── High Revenue | Probability: 0.4647 | Payoff: 537.01
  └── Low Revenue  | Probability: 0.5353 | Payoff: 130.92
  Expected Value: 319.63
```

```
 Home and lifestyle
  ├── High Revenue | Probability: 0.5125 | Payoff: 529.58
  └── Low Revenue  | Probability: 0.4875 | Payoff: 133.8
  Expected Value: 336.64

 Sports and travel
  ├── High Revenue | Probability: 0.5301 | Payoff: 518.55
  └── Low Revenue  | Probability: 0.4699 | Payoff: 121.68
  Expected Value: 332.07

Q37. Business Recommendation:
The recommended product line is Home and lifestyle because it has the highest Expected Value of 336.64 .
```

Conclusion:

The Decision Making under Uncertainty practical successfully analyzed the supermarket sales dataset using decision tree concepts and Expected Value. The revenue of each product line was classified into High Revenue and Low Revenue based on the median revenue, and the probabilities and payoffs of the outcomes were calculated. Expected Value was then used to compare all product lines. The analysis showed that Home and lifestyle has the highest Expected Value and is therefore the best product line to select for the business decision. This practical demonstrates how probability, payoff, and Expected Value can be used to make effective business decisions under uncertainty.